

DANIEL NG

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Education

University of British Columbia

BASc in Electrical Engineering (Co-op, available May 2026)

Expected Graduation: May 2028

Vancouver, BC

Experience

Norsat International Inc.

Hardware Intern

June 2025 – August 2025

Richmond, BC

- Characterized Ku-Band Block-Up Converters; measured gain, power saturation, 1-dB compression point, spurious emissions, IMD3, and noise figure using Vector Network Analyzers, Signal Analyzers, oscilloscopes, and multimeters to verify hardware parameters.
- Swept DAC codes to calibrate RF power modules to monitor and optimize output power and stability before testing.
- Assembled and troubleshooted BUCs and LNAs at the PCBA level, performing microscopic SMT and THT soldering and microscopic inspection on MMIC chips to meet production quality standards.
- Verified and troubleshooted SSPA boards by conducting continuity tests and cross-referencing schematics to identify and rectify DC bias instability and RF path discontinuities, ensuring 100% yield for high-power MMIC stages.

Formula UBC (Electrical/Firmware)

Project Lead: Timing Gates | *Embedded C, Altium, Schematic and Layout Design*

September 2025 – Present

- End-to-end development of a laser receiver with two MechE students to develop optomechanical shrouding coupled with a custom 3V3 CMOS analog front end and 2.4GHz communication between master-slave ESP32-S3 MCUs to calculate speed and lap times.
- Characterizing baseline photodiode thermal dark current and TIA voltage noise to calibrate an adjustable hysteresis threshold via potentiometer, targeting a 20dB Signal-to-Noise Ratio (SNR) and a Bit Error Rate (BER) of $< 10^{-7}$ to ensure reliable high-speed data integrity between MCUs.

Projects

RISC-V Simple CPU | *SystemVerilog, Altera DE10-Lite FPGA, Quartus Prime, Questa*

- Architected and implemented a Single-cycle RISC-V processor in SystemVerilog with memory-mapped I/O onto Altera's DE10-Lite with functioning register file, ALU, and control unit, supporting a 10-instruction subset of the RISC-V ISA.
- Wrote a self-checking non-synthesizable testbench with assertion-based verification to verify instruction-level correctness and used Questa to trace signals through the synchronous timing datapath.

Tron Legacy VGA Driver | *C, Altera DE10-Lite FPGA, MMIO, VGA*

- Developed a custom VGA graphics driver using Memory-Mapped I/O (MMIO) to manage frame buffers, rendering dynamic light traces and a gameplay perimeter while maintaining high-speed pixel updates.
- Implemented an interrupt-driven control scheme for human input, utilizing an Interrupt Service Routine (ISR) to handle asynchronous pushbutton events, significantly reducing input latency compared to traditional polling.
- Engineered an autonomous robot opponent with predictive collision logic; the algorithm performs real-time memory reads of the VGA buffer to scan for obstacles and dynamically updates movement vectors to avoid crashes.

Interrupt-Driven Alarm Clock System | *A51 Assembly, N76E003 (8051) MCU, ISRs, CrossIDE*

- Engineered a real-time settable clock using a 16.6 MHz oscillator to drive timers and asynchronous interrupts.
- Utilized Timer 0 to generate a stable 2 kHz square wave for audio signaling and Timer 2 to manage high-precision half-second BCD clock increments.
- Interfaced a 4-bit LCD for time/alarm display and implemented a software debouncing "wait-and-see" algorithm coupled with decoupling capacitors for pushbutton inputs to eliminate noise.

Technical Skills

Digital Design/HDL/Low-Level: SystemVerilog, RISC-V Assembly, A51 Assembly, FSM Implementation & Design

Embedded Systems: CAN, USB, SPI, UART, I2C, ESP32 and 8051-family MCUs, C Embedded Firmware Development, RV32 Bare Metal Programming, MMIO

Design Tools: Quartus Prime, Questa/ModelSim, Altium Designer, CPULator

Hardware & Lab: VNA, Signal Analyzer, PCB Design and Rework, SMT/THT Soldering, Oscilloscope, Multimeter